Patch- and Class-Wise Hyperspectral Knowledge Learning: A Composite Consistency-Constrained Self-Ensemble Framework for Change Detection

Xiaoyang Zhao, Siyao Li, Xinyue Liu, Chuanming Song, and Xianghai Wang

***Abstract*— Obtaining fine land surface change information from multitemporal hyperspectral images (HSIs) is a key goal pursued in remote sensing image processing. Recently, HSI change detection (HSI-CD) methods based on convolutional neural networks (CNNs) have achieved surprising detection results. One of the reasons is the support of large-scale labeled samples for network learning. However, the existence of mixed pixels greatly increases the difficulty of HSI interpretation, resulting in accurate pixel-level labeling work with a heavy burden and unable to meet the needs of time-sensitive applications. For this reason, achieving stable and high-precision CD with fewer samples is a difficult issue in this field. To address the above problems, a composite consistency-constrained self-ensemble framework (C3SelF) for HSI-CD is proposed, to alleviate the problems of low detection accuracy and instability caused by small samples. The framework mainly comprises two lightweight networks with the same structure aiming at accelerating the model inference process and thus improving the processing timeliness. The composite learning mode implements patch-wise classification loss, class-wise consistency loss on labeled samples, and patch-wise consistency loss on unlabeled samples under a multilevel noise perturbation strategy, which improves the classification results and reduces the labeling cost. Moreover, to exploit the multidimensional features contained in HSIs, a lightweight selective spatial–spectral feature joint network (S3Net) is designed to overcome over-fitting, and to deeply mine the discriminative information in unlabeled samples, a new sample screening strategy is designed to ensure the stability of the network during training unlabeled samples. Extensive experiments prove that the proposed C3SelF outperforms the state-of-the-art (SOTA) methods at a sampling rate of 0.1%, reaching 93.44% Kappa and 97.26% overall accuracy (OA) on the Farmland dataset. The source code of the proposed framework will be released at https://github.com/zxylnnu/C3SelF.**

***Index Terms*— Change detection (CD), convolutional neural networks (CNNs), deep learning, hyperspectral image (HSI) processing, semi-supervised learning.**

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I. Introduction

As A hot issue in the field of remote sensing image processing, hyperspectral image change detection (HSI-CD) aims to provide timely and meaningful accurate land surface change information for practical applications such as precision agriculture, natural disaster assessment, and resource extraction by analyzing multitemporal HSIs acquired at the same geographic location [1].

Early HSI-CD methods mainly analyze change information by directly calculating and comparing the pixel values and simple features of bi-temporal images. For example, Ertürk et al. [2] organically combine change vector analysis with spectral angle mapper and propose a fuzzy inference combination strategy to measure the changing intensity in terms of both angle and magnitude; Marinelli et al. [3] propose a binary spectral change vector method that can convert real-valued spectral change vectors into binary form for detection in discrete high-dimensional space. However, due to the Hughes phenomenon of HSIs, when the spectral characteristics are seriously disturbed and distorted, it will have a certain degree of negative impact on learning discriminative features. Some methods based on dimensionality reduction have been developed, and the dimensionality reduction-based methods can be divided into two types: band selection-based methods and spectral feature extraction-based methods. The former selects representative bands to represent the original data. After comprehensively analyzing the impact of band selection on HSI-CD, Liu et al. [4] replace hyperspectral data space with a series of bands with the most information to avoid Hughes. Whereas the latter retains important information through mathematical transformation methods that project high-dimensional data into a low-dimensional feature space, it is usually used as a means of feature preprocessing. Wu et al. [5] propose a multitemporal principal component analysis (PCA) method to reduce HSI dimensionality, and use the slow feature analysis method on low-dimensional data to increase the difference between changed and unchanged features. Furthermore, to reduce the interference caused by the mixed pixel problem, some spectral unmixing-based methods

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have been proposed. Such as Jafarzadeh and Hasanlou [6] distinguish between changed and unchanged information based on the estimation of the endmembers.

Recently, with the significant improvement of hardware computing and large-scale collection of remote sensing data, deep learning technology has been widely used in many fields, and HSI-CD technology based on deep learning also developed rapidly. Ding and Li [7] integrate the attention mechanism (AM) with recurrent neural networks (RNNs) to obtain spatial and spectral feature representations. To overcome the problem of insignificant feature representation due to early or late fusion strategies, Qu et al. [8] propose a multilevel encoder–decoder attention network. In addition to the above methods that use RNNs to analyze temporal correlation, existing methods use convolutional neural networks (CNNs) more. To fully exploit multilevel features, Luo et al. [9] propose a multiscale differential change feature fusion network to capture the subtle differences between bi-temporal HSIs at each scale through a bidirectional differential changed feature module.

Considering these problems, some strategies have been developed to alleviate the detection limitations brought about by a small number of samples. The existing methods to solve this problem can be roughly divided into two types. One is to increase the size of the training dataset. Generate more labeled samples through scaling, rotation, and flipping, to increase the amount of the learnable features of the network, and reduce the risk of overfitting; The other is to enhance the generalization of the network under fewer samples by some specific learning strategies, such as self-supervised learning and unsupervised learning. Tang et al. [10] use a pretrained autoencoder with multiscale 3-D convolutional blocks to comprehensively extract spatial–spectral joint features; Wang et al. [11] introduce few-shot learning strategy into this research field, and propose a cross-heterogeneous domain framework with bidirectional generation model; Tang et al. [12] propose an unsupervised method that combines graph convolution network and metric learning to help network training by generating reliable pseudo-labels;

Proving that the problem of few-shot has received widespread attention in HSI-CD. Although data augmentation techniques can increase the number of learnable features of the network by generating more labeled samples, this approach is still limited by the quantity and quality of the original labeled data. Meanwhile, although self-supervised and unsupervised learning methods can improve the generalization ability of the network to some extent, they do not make full use of the existing label information, resulting in limited detection accuracy. Pseudo-label-based methods rely on the prediction of unlabeled data to generate pseudo-labels, which are then used for network training. However, there is often uncertainty in the assignment of pseudo-labels, which may cause the network to learn the wrong feature representation, thus affecting the final detection accuracy. Few-shot learning-based methods usually need to consider the problem of domain alignment when dealing with cross-domain data. Differences in data distribution between different domains may lead to performance degradation of the network during migration learning, especially in the HSI-CD task, where the domain alignment problem is particularly prominent due to the complexity and diversity of remote sensing images. For this reason, there remains still a great deal of work to do for further research. In this article, a composite consistency-constrained self-ensemble framework (C3SelF) is proposed, which can achieve stable and high-precision detection with fewer samples through semi-supervised learning (SSL) mode. The main contributions are given below in detail.

1) A C3SelF that combines patch-wise knowledge learning with class-wise knowledge learning is proposed, and a multilevel noise perturbation strategy is introduced to enable the network to maintain stable and high-precision detection under fewer samples.
2) A lightweight selective spatial–spectral feature joint network (S3Net) is designed, which is with a spatial selective attention module and a spectral selective attention module so that the network can dynamically and adaptively select the salient features to be concerned. This automated feature selection mechanism helps improve detection accuracy and reduces the dependence on manual design feature extraction strategies.
3) A matrix for calculating class-wise knowledge is designed, which represents the amount of information transferred within and across classes. This matrix enables consistency constraints on class-wise label content during the ensemble learning process on the basis of obtaining the contribution of the current sample to each class. This helps to maintain the stability of the network prediction and improves the detection performance under fewer sample conditions.
4) To utilize the effective information in unlabeled samples, a metric-based sample screening strategy is designed. The available sample information contained in unlabeled samples is twofold considered in terms of predictive stability and confidence. The unlabeled samples are filtered to further enrich the training data and improve the generalization ability of the network.

The rest is arranged as follows. Section II focuses on the application of the semi-supervised learning strategy used in HSI-CD, as well as the research status of various AMs applied in this field. Section III provides a detailed process of C3SelF. Section IV demonstrates the detection of each dataset at 0.1% sampling rate. The conclusion is made in Section V.

II. Related Works

A. Semi-Supervised Learning Used in HSI-CD

Due to factors such as the relatively high cost of obtaining sample labels, scarcity of labeled data, or imbalanced data distribution, traditional fully supervised learning models tend to exhibit only limited detection performance when facing these challenges. In this context, the SSL strategy has merged [13]. SSL is a commonly used machine learning strategy that aims to use unlabeled data with a small amount of labeled data to train the network together.

By exploring the potential distribution and characteristics embedded in the samples, the network can adapt to complex

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data more stably, so that the generalization ability and robustness can be enhanced, and achieve the purpose of alleviating the large demand for high-quality labeled samples in fully supervised networks. Recently, graph-based SSL has become a hot research topic in this field. It can break through the limitation that CNNs can only model Euclidean data and extract the topological structure in the image. The labels of labeled data are propagated to unlabeled data through edges established by the similarity measurement between nodes to achieve the SSL mode of the network. Wang et al. [14] designed a semi-supervised domain-aligned transformer network with a two-branch transformer self-encoder to align the features of dual-temporal hyperspectral images (HSIs) by cyclic training to reduce the detection of interfering information due to cross-domain.

Wang et al. [15] employed graph attention and proposed a dual-branch framework with a complementary strategy to obtain multidimensional features by constructing the joint temporal–spatial graph attention matrix. Dong et al. [16] and Zhao et al. [17] used the graph transformer to capture change features, where the former used it to flexibly construct the global information necessary to distinguish complex objects. The latter uses the SSL strategy of graph convolution along with a graph transformer to combine local and global features organically. In general, this type of method reduces the need for labeled samples by constructing an adjacency matrix and propagating labels to connected nodes that have homogeneous features.

The SSL method based on pseudo-labels trains the network using labels first, and, afterward, unlabeled data is predicted by this model. According to a sample query function, the predictions with higher confidence are labeled, which are then added to the training dataset. Expand the number of training samples and repeatedly train the network, the above-mentioned process is iterative until the stopping condition is satisfied. Yu et al. [18] proposed a semi-supervised target detection method for HSIs, which achieves the best current performance by choosing a suitable data enhancement method and pseudo-label generation strategy. To reduce the dependence on labeled samples while maintaining high detection accuracy, Chen et al. [19] proposed an active learning strategy, aiming at selecting high-value labeled samples from the test dataset as additional training data.

Graph-based methods tend to use superpixels as the analysis unit because constructing a pixel-level structured graph will face a huge computational load. This operation leads to two shortcomings: first, it makes the detection dependent on the accuracy of superpixel segmentation, and second, it is not able to effectively capture the scatter-shape change region. The accuracy of the pseudo-label-based methods heavily depends on the design of the sample query function, but the network is not robust enough when faced with class imbalance and abnormal samples.

Some methods employ semi-supervised learning strategy based on self-training [20], mean-teacher [21], and generative adversarial models [22], but there is a lack of exploration in the field of HSI-CD.

B. AMs Applied in HSI-CD

AMs can help the network quickly focus on key features that need to be learned, thus significantly improving the network performance and efficiency, and have been used on a large scale. Since various features contribute differently to network learning, the AMs commonly used in this field are channel attention, spatial attention, and temporal attention. In particular, channel attention allows the network to adaptively select which features are effective, spatial attention enables the network to pay attention to the location of salient features in the input, and temporal attention helps the network to analyze which temporal features are more beneficial for learning.

As an effective module, the convolutional block attention module (CBAM) [23] is widely used in various methods. Ding and Li [7] used it to obtain spatial and spectral feature representation. In addition to CBAM, methods based on Self-AM [24] are also proposed. Wang et al. [25] used it to effectively explore spatial information.

On the basis of Self-AM, transformer structure based on multihead Self-AM has also been widely explored. To utilize mid- and long-range information, Wang et al. [26] proposed a triplet branch transformer framework with parent- and brother-AM to explore the influence of bi-temporal images on the differential image. Taking long-range correlation between bands into account, Zhang et al. [27] implemented a transformer module along the spectral dimension to increase the difference between spectral features.

Overall, the AM can be widely used as an effective means to enable the network to strengthen the learning of important information and suppress the interference of invalid information, greatly improving the detection performance in this field. However, the research on AMs in this field is still in its infancy, and most of the networks only introduce and use the original module. Further exploration is needed on how to improve the attention module to efficiently capture salient features from the input.

III. Methodology

A. Overall Structure of C3SelF

Fig. 1 shows the overall flowchart of C3SelF. It mainly includes a CD network and an ensemble learning network. They share the same network structure. During the training process, the CD network (f^D) performs patch-wise supervised learning on labeled samples with single-level noise added; then, to enhance the robustness of the network under fewer samples, f^D and the ensemble learning network (f^E) need to perform class-wise unsupervised consistency learning on labeled samples with single-level noise and labeled samples with multilevel noises, respectively; finally, to exploit the valid information embedded in unlabeled samples, f^D and f^E need to perform patch-wise unsupervised consistency learning on unlabeled samples with single-level noise and labeled samples with multilevel noises, respectively.

During each iteration, f^D is constrained using the triple loss described above, and its parameters are updated using the Adam algorithm. The parameter update mode of f^E

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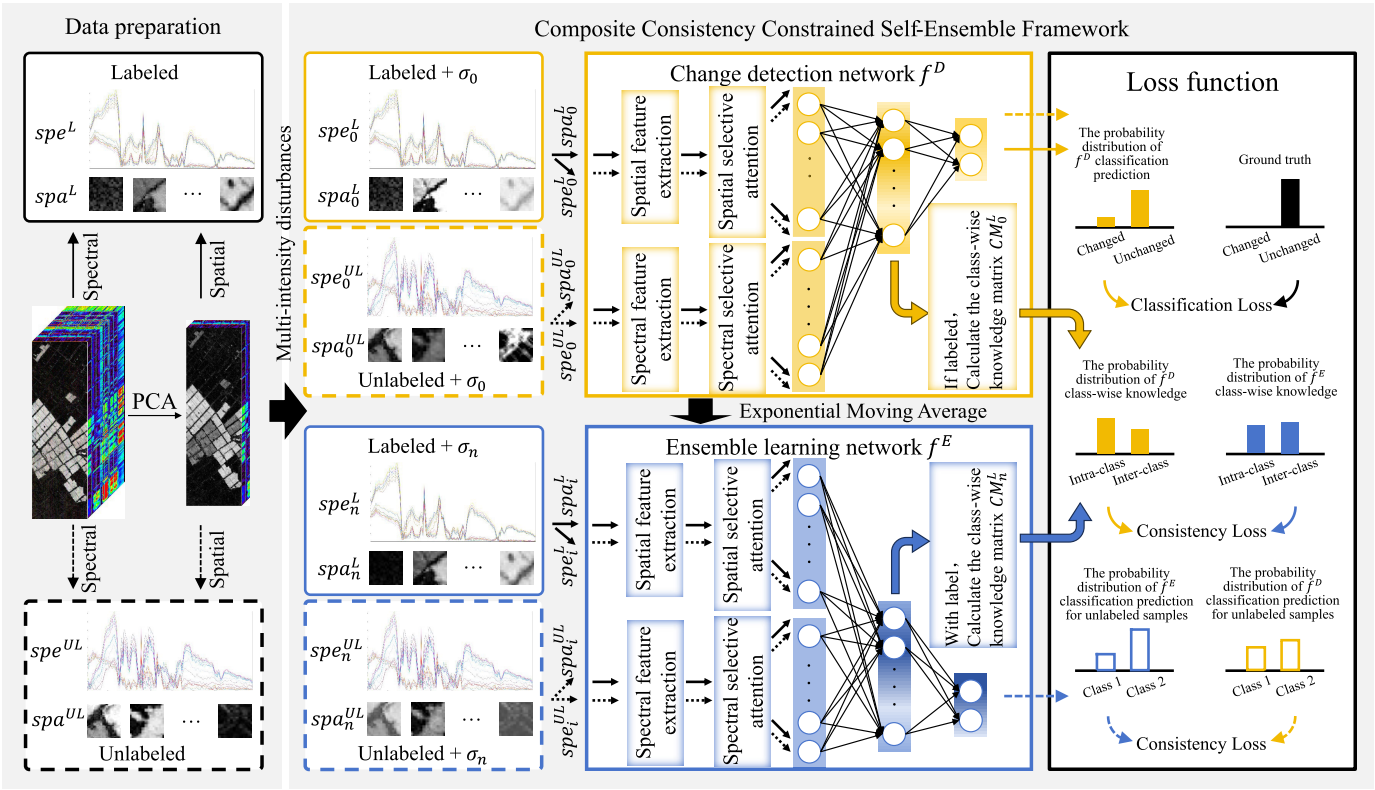


Fig. 1. Flowchart of C3SelF. During training, the spectral and spatial features of labeled and unlabeled HSIs are extracted (solid arrows indicate the processing flow for labeled samples and dashed arrows indicate the processing flow for unlabeled samples), and the data are fed into the CD network, and the ensemble learning network after being added by single-level and multilevel noise, respectively. The CD network performs supervised learning on labeled samples, and the two networks perform unsupervised consistency learning on noise-added samples patch-wise and class-wise, respectively.

is different from that of f^D. It uses the exponential moving average to fuse the current parameters of f^D and the parameters of the previous iteration of f^E to achieve the acquisition of multiple effective features. After finishing the training process, f^E is applied to predict test data, and the change map is obtained. The flow of C3SelF is given in Algorithm 1.

Since f^D and f^E have the same network structure, the following will take f^D as an example to describe specifically, as well as the loss function and optimization.

Considering that an overly deep network structure will bring about serious overfitting and reduce the generalization ability of the network under the few-shot condition, to this end, a lightweight S3Net is designed to extract spatial high-response features and spectral high-response features from the input efficiently.

Define bi-temporal HSIs as $X_1 \in R^{H\times W\times B}$ and $X_2 \in R^{H\times W\times B}$, the differential image as $X \in R^{H\times W\times B}$, where H, W, and B are height, width, and number of bands, respectively. First, obtain the spectral data $X_{\text{spe}} \in R^{HW\times B}$; then, use the PCA algorithm to reduce the dimension of X, and only the first k ($k = 5$) principal components are retained. The dimensionality reduction data is denoted as $X' \in R^{H\times W\times 5}$, so that the spatial data X_{spa} of X can be obtained using ws × ws as the window size, denoting $X_{\text{spa}} \in R^{HW\times \text{ws}\times \text{ws}\times 5}$.

As shown in Fig. 2, the framework has two branches, named the spectral branch and the spatial branch. Each branch has a feature extraction module and a selective attention module. Among them, the spatial feature extraction module relies on the 2-D convolutional layer, ReLU activation layer, average pooling layer, and skip connection to extract coarse spatial features $\text{Fc}_{\text{spa}} \in R^{64\times(H/4)\times(W/4)}$. In the spectral feature extraction module, there is only one linear layer and ReLU activation layer to extract coarse spectral features $\text{Fc}_{\text{spe}} \in R^{1024}$. Next, Fc_{spa} and Fc_{spe} are refined through the spatial selective AM (SpaSAM) and the spectral selective AM (SpeSAM), respectively.

B. Network Structure of f^D

SpaSAM first aggregates spatial response feature maps with different response levels by averaging and maximizing each spatial position vector of Fc_{spa}, obtaining the spatial average response feature map $F_{\text{spa}}^{\text{avg}} \in R^{1\times(H/4)\times(W/4)}$ and spatial maximum response feature map $F_{\text{spa}}^{\max} \in R^{1\times(H/4)\times(W/4)}$; subsequently, the spatial attention is activated on the concatenated spatial response feature maps using 2-D convolution operation followed by Sigmoid function, the spatial average attention map $\text{att}_{\text{spa}}^{\text{avg}} \in R^{1\times(H/4)\times(W/4)}$, and spatial maximum attention map $\text{att}_{\text{spa}}^{\max} \in R^{1\times(H/4)\times(W/4)}$ are acquired to capture the spatial position information to be enhanced or suppressed under dual-level responses; next, the weights $w_{\text{spa}}^{\text{avg}} \in R^{1\times(H/4)\times(W/4)}$ for $\text{att}_{\text{spa}}^{\text{avg}}$ and $w_{\text{spa}}^{\max} \in R^{1\times(H/4)\times(W/4)}$ for $\text{att}_{\text{spa}}^{\max}$ can be obtained by Softmax function, and the fused spatial selective attention map $\text{fuatt}_{\text{spa}} \in R^{1\times(H/4)\times(W/4)}$ is achieved after weighting; finally, the fine spatial feature map $\text{Ff}_{\text{spa}} \in R^{64\times(H/4)\times(W/4)}$ that

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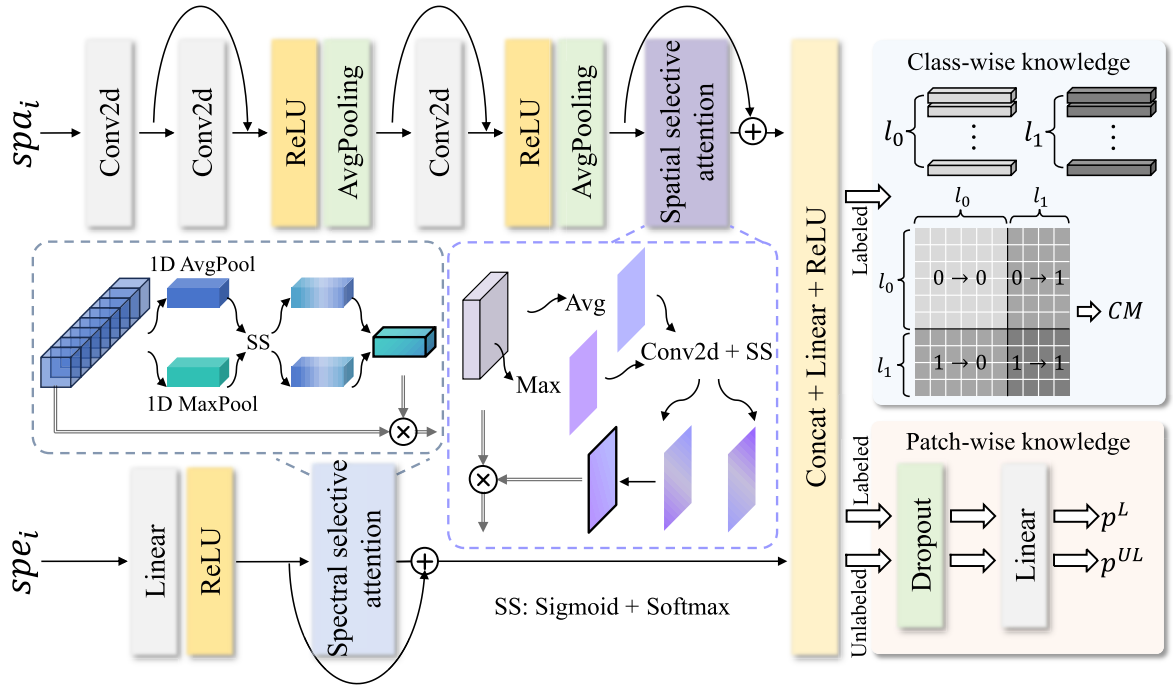


Fig. 2. Schematic of the S3Net. The spatial and spectral features go through the spatial attention module and the spectral attention module, respectively. Patch-wise prediction is performed for unlabeled samples, and patch-wise prediction and class-wise metrics are learned for labeled samples.

Algorithm 1 Pipeline of C3SelF

Input: Labeled samples: x^L (including spe^L and spa^L), labels: y^L, unlabeled samples: x^{UL} (including spe^{UL} and spa^{UL}), multilevel noises: $\sigma_{0,1,2,3}$, epoch: Ep;

for $e = 1$ to Ep **do**
- Add σ_0-level Gaussian noise to x^L and x^{UL}, denoted as x_0^L and x_0^{UL};
- The prediction of x_0^L by f^D is denoted as $\hat{y}^L$, and calculate the CM_0^L of x_0^L;
- Calculate the binary cross entropy loss L_{bce} between $\hat{y}^L$ and y^L according to Eq. (4);
- The prediction of x_0^{UL} by f^D is denoted as $\hat{y}_0^{UL}$;
- Add $\sigma_{1,2,3}$-level Gaussian noises to x^L and x^{UL}, denoted as $x_{1,2,3}^L$ and $x_{1,2,3}^{UL}$;
- Calculate the $CM_{1,2,3}^L$ of $x_{1,2,3}^L$ by f^E;
- Calculate the class-wise consistency loss L_{cls} between CM_0^L and $CM_{1,2,3}^L$ according to Eq. (5);
- The prediction of $x_{1,2,3}^{UL}$ by f^E is denoted as $\hat{y}_{1,2,3}^{UL}$;
- Screen unlabeled samples according to Eq. (6) and Eq. (7);
- Calculate the patch-wise consistency loss L_{cons} between $\hat{y}_0^{UL}$ and $\hat{y}_{1,2,3}^{UL}$ according to Eq. (8);
- Calculate the total loss L_{sum} of f^D according to Eq. (9);
- Back propagate L_sum, and use the Adam algorithm to update the parameters of f^D;
- Update the parameters of f^E according to Eq. (11);

end for

containing spatial selective attention is obtained by multiplying with $\mathrm{Fc_{spa}}$. In summary, SpaSAM can be expressed by the following equation:

$$\begin{cases} \mathrm{att_{spa}} = \sigma\left(C\left(\mathrm{cat}\left(\mathrm{mean}\left(\mathrm{Fc_{spa}}\right), \max\left(\mathrm{Fc_{spa}}\right)\right)\right)\right) \\ w_{\mathrm{spa}} = \mathrm{Softmax}\left(\mathrm{att_{spa}}\right) \\ \mathrm{fuatt_{spa}} = w_{\mathrm{spa}}^{\mathrm{avg}} \times \mathrm{att_{spa}^{avg}} + w_{\mathrm{spa}}^{\max} \times \mathrm{att_{spa}^{max}} \\ \mathrm{Ff_{spa}} = \mathrm{fuatt_{spa}} \times \mathrm{Fc_{spa}} \end{cases} \tag{1}$$

where $\mathrm{cat}(\cdot,\cdot)$ represents the concatenation operation, $\sigma(\cdot)$ represents the Sigmoid function, and $C(\cdot)$ represents a 2-D convolution operation with two kernels of size 1×1.

SpeSAM first aggregates spectral response feature vectors with different response levels by average pooling and maximum pooling for the input feature vector $\mathrm{Fc_{spe}}$, which are namely spectral average response feature vector $F_{\mathrm{spe}}^{\mathrm{avg}} \in R^{1024}$ and spectral maximum response feature vector $F_{\mathrm{spe}}^{\max} \in R^{1024}$. Afterward, the spectral attention activation is performed on the concatenated spectral response feature vectors using the Sigmoid function to obtain the spectral average attention vector $\mathrm{att_{spe}^{avg}} \in R^{1024}$ and spectral maximum attention vector $\mathrm{att_{spe}^{max}} \in R^{1024}$, respectively, to capture the spectral information that should be strengthened or suppressed under different responses; next, the weights $w_{\mathrm{spe}}^{\mathrm{avg}} \in R^{1024}$ for $\mathrm{att_{spe}^{avg}}$ and $w_{\mathrm{spe}}^{\max} \in R^{1024}$ for $\mathrm{att_{spe}^{max}}$ are obtained by Softmax function, and the fused spectral selective attention vector $\mathrm{fuatt_{spe}} \in R^{1024}$ is achieved after weighting; finally, by multiplying the input $\mathrm{Fc_{spe}}$, the fine spectral feature vector $\mathrm{Ff_{spe}} \in R^{1024}$ for obtaining spectral selective attention is obtained. In summary, the SpeSAM can be expressed by the following equation:

$$\begin{cases} \mathrm{att_{spe}} = \sigma\left(\mathrm{cat}\left(\mathrm{AP}\left(\mathrm{Fc_{spe}}\right), \mathrm{MP}\left(\mathrm{Fc_{spe}}\right)\right)\right) \\ w_{\mathrm{spe}} = \mathrm{Softmax}\left(\mathrm{att_{spe}}\right) \\ \mathrm{fuatt_{spe}} = w_{\mathrm{spe}}^{\mathrm{avg}} \times \mathrm{att_{spe}^{avg}} + w_{\mathrm{spe}}^{\max} \times \mathrm{att_{spe}^{max}} \\ \mathrm{Ff_{spe}} = \mathrm{fuatt_{spe}} \times \mathrm{Fc_{spe}} \end{cases} \tag{2}$$

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where AP(·) represents 1-D average pooling and MP(·) represents 1-D max pooling.

After extracting the fine spatial feature map $\mathrm{Ff}_{\mathrm{spa}}$ and the fine spectral feature vector $\mathrm{Ff}_{\mathrm{spe}}$, first perform dimension transformation on $\mathrm{Ff}_{\mathrm{spa}}$ and then combine it with $\mathrm{Ff}_{\mathrm{spe}}$ along the channel dimension; afterward, the spatial–spectral joint feature map F_{ss} is extracted through a linear layer followed by the ReLU activation function. Finally, if the input data belongs to unlabeled data, the dropout operation, linear layer, and Softmax function are used to predict the patch-wise change probability; if the input data belongs to labeled data, in addition to predicting the patch-wise change probability, it is also necessary to calculate the class-wise knowledge matrix CM, which represents the amount of information transferred within the intraclass and across the interclass and is used to further supervise the learning between f^D and f^E.

Denote the batch size as Bs, there are l_k samples of class k $(k = 0, 1)$ in the current batch, $\sum_k l_k = Bs$. The calculation of CM is as shown in (3). It first calculates the contribution of the current sample to other samples. And then, calculate the stability of the sample, that is, the amount of information currently transferred within the intraclass and across the interclass. The higher the stability, the higher the probability of intraclass information transfer and the lower the probability of interclass information transfer

$$\mathrm{CM}^{a,c} = \mathrm{sum}\left(\mathrm{std}\left(\mathrm{Softmax}\left(F_{\mathrm{ss}}^{a,c} \times \left(F_{\mathrm{ss}}^{a,c}\right)^T\right)\right)\right) \tag{3}$$

where $\{(a, c)|(0, 0), (0, 1), (1, 0), (1, 1)\}$, std(·) represents the standard deviation calculation, and sum(·) represents the summation. If $a = c$, $\mathrm{CM}^{a,c}$ means the information transferred within the class; else $\mathrm{CM}^{a,c}$ means the information transferred between the classes.

C. Loss Function and Optimization Strategies

Define $\mathbb{X}^L = \{x^L[i]\}_{i=1}^{n_L}$ and $\mathbb{X}^{\mathrm{UL}} = \{x^{\mathrm{UL}}[i]\}_{i=1}^{n_{\mathrm{UL}}}$ to represent the set of labeled samples and the set of unlabeled samples, where n_L is the number of labeled samples and n_{UL} is the number of unlabeled samples. $\mathbb{Y}^L = \{y^L[i]\}_{i=1}^{n_L}$ is the label corresponding to $\mathbb{X}^L$, then the first classification supervision loss L_{bce} of the f^D can be defined in terms of the binary cross entropy loss

$$L_{\mathrm{bce}} = -\sum_{m=1}^{Bs}\left(y_m^L \log \hat{y}_{0m}^L + \left(1 - y_m^L\right)\log\left(1 - \hat{y}_{0m}^L\right)\right) \tag{4}$$

where Bs is the batch size and $\hat{y}_{0m}^L$ is the prediction of f^D for the mth sample in x_0^L.

Considering the property that the amount of information transferred within the intraclass and across the interclass should remain stable under multi-intensity perturbations, and thus class-wise consistency supervision on f^D can be defined as

$$L_{\mathrm{cls}} = \sum_{n=1}^{3}\sum_{m=1}^{Bs}\left\|\mathrm{CM}_{0\,m}^L - \mathrm{CM}_{n\,m}^L\right\|^2 \tag{5}$$

where n is the degree of noises, $\mathrm{CM}_{0\,m}^L$ is the class-wise knowledge matrix calculated by f^D for the mth sample in x_0^L, and $\mathrm{CM}_{0\,m}^L$ is the class-wise knowledge matrix calculated by f^E for the mth sample in x_n^L.

To take advantage of the information contained in unlabeled samples, patch-wise consistency supervision L_{cons} is performed on f^D. However, most of the unlabeled predictions of the network are not accurate enough in the initial learning phase. If all unlabeled samples are used to participate in the training, it will undoubtedly increase the learning difficulty of the network and affect the convergence speed. On this basis, a screening principle for unlabeled samples is proposed, aiming to identify unlabeled samples with higher confidence for calculating L_{cons}.

After f^E predicts $\hat{y}_n^{\mathrm{UL}}$, the prediction needs to be analyzed. Refer to the consistency of prediction, the larger the consistency P_m^{uc}, the smaller the standard deviation, which means that the prediction of $\hat{y}_n^{\mathrm{UL}}$ by f^E is consistent. Refer to the probability of prediction, the higher the value P_m^{pro}, the higher the probability that f^E predicts x_n^{UL} to belong to this class

$$\begin{aligned} P_m &= P_m^{\mathrm{uc}} + P_m^{\mathrm{pro}} \\ &= -\sum_{n=1}^{3}\mathrm{std}\left(\hat{y}_n^{\mathrm{UL}}(m)\right) + \sum_{n=1}^{3}\mathrm{mean}\left(\hat{y}_n^{\mathrm{UL}}(m)\right) \end{aligned} \tag{6}$$

where mean(·) represents the mean calculation. Finally, P_m is arranged in descending order from largest to smallest, and the first sn samples are used to calculate L_{cons}

$$sn = \mathrm{int}\left(e^{-\left(1-\frac{t}{Ep}\right)^2} Bs + 0.999\right) \tag{7}$$

where int(·) is the rounding function and t is the current epoch.

L_{cons} can be defined as

$$L_{\mathrm{cons}} = \sum_{n=1}^{3}\sum_{m=1}^{sn}\left\|\hat{y}_{0m}^{\mathrm{UL}} - \hat{y}_{n_m}^{\mathrm{UL}}\right\|^2 \tag{8}$$

where n is the degree of noises, $\hat{y}_{0m}^{\mathrm{UL}}$ is the prediction of f^D for the mth sample in x_0^{UL}, and $\hat{y}_{n_m}^{\mathrm{UL}}$ is the prediction of f^E for the mth sample in x_n^{UL}.

In summary, the total loss function L_{sum} can be defined as

$$L_{\mathrm{sum}} = L_{\mathrm{bce}} + L_{\mathrm{cls}} + L_{\mathrm{cons}}. \tag{9}$$

According to the learning mode of self-ensemble models, the optimization of C3SelF consists of two parts. The first is the optimization of f^D, which can be represented by the following objective function. It is solved by the Adam algorithm

$$\min_{f^D}(L_{\mathrm{sum}}). \tag{10}$$

Refer to f^E, after updating f^D in each epoch, use the exponential moving average strategy to update the parameters in f^E according to the following equation:

$$\begin{cases} \Theta_t^E = \lambda(t)\Theta_{t-1}^E + (1 - \lambda(t))\Theta_t^D \\ \lambda(t) = 1 \times e^{\left(-5\left(1-\frac{t}{Ep}\right)^2\right)} \end{cases} \tag{11}$$

where $\lambda(\cdot)$ is the sigmoid ramp-up function, Θ_t^E represents the parameters of f^E under the tth epoch, Θ_t^D represents the parameters of f^D under the tth epoch, and Ep is the total number of epochs.

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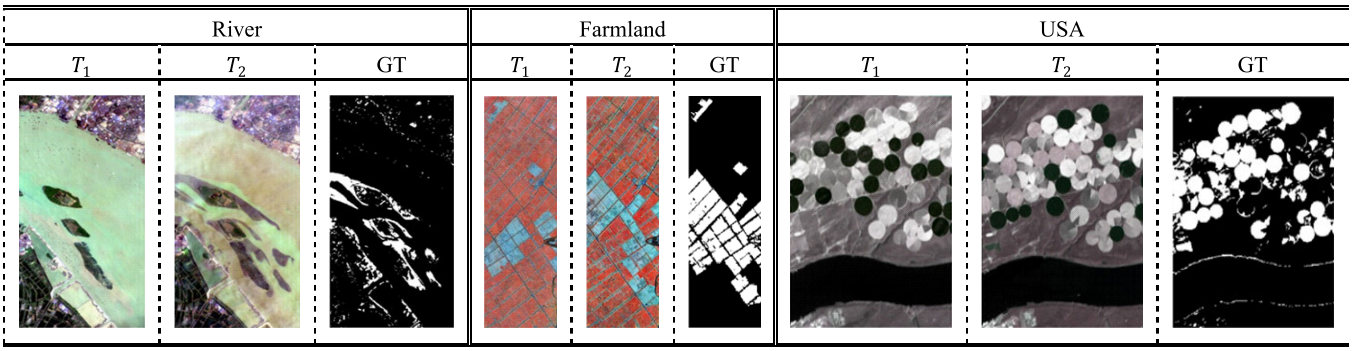


Fig. 3. Pseudo-color visualization of the datasets.

TABLE I
DATASET DETAILS

Dataset	River	Farmland	USA
Size	463×241	450×140	307×241
Band	198	155	154
Date T_1	May 3rd, 2013	May 3rd,2006	May 1st,2004
Date T_2	Dec 31st, 2013	Apr 23rd,2007	May 8th,2007
Sampling rate	0.1%	0.1%	0.1%
Labeled samples	112	64	74
Unlabeled samples	111471	62936	73913

IV. EXPERIMENTS AND ANALYSIS

To validate the effectiveness and excellent detection performance of C3SelF under fewer samples, experiments with a sampling rate of 0.1% have been conducted on three commonly used and publicly available HSI-CD datasets.

A. Datasets and Experimental Setup

In this article, the detection performance of the proposed C3SelF is evaluated using the following three HSI-CD datasets: River [28], Farmland [29], and USA [30]. There are bi-temporal HSIs and the corresponding ground-truth (GT) map in each dataset. Table I provides detailed information about these three datasets, including basic information, and the number of samples at the sampling rate of 0.1%. Fig. 3 shows the pseudo-color visualization.

To verify the superiority of C3SelF, we compared it with six methods. The details of these compared methods are as follows.

1) *CSDBF [21]:* It adopts a two-branch structure to extract superpixel-level and pixel-level features, respectively, and strengthens the feature discriminative property by an attention module.
2) *DA-Former [14]:* By designing a two-branch temporal autoencoder with shared intermediate layer weights, features from different data are pulled into the same space, and the domain is aligned using a loop training method.
3) *CNCMN [31]:* By combining image convolution and graph convolution, it effectively learns Euclidean and non-Euclidean neighbor information of target pixels in HSIs for efficient feature representation and classification.
4) *MSDFFN [13]:* A temporal feature encoder–decoder subnetwork is designed to extract temporal features from HSIs.
5) *GTMSiam [32]:* Through multiscale feature extraction and gating information transfer, it realizes the reduction of model complexity while ensuring detection accuracy.
6) *ML-EDAN [10]:* By making full use of hierarchical features through a multilevel fusion strategy, it realizes effective transformation of spatial–temporal features and temporal dependency analysis.

The experiments are implemented under the PyTorch 1.12.10 framework, an NVIDIA Titan X (Pascal) GPU is used for training, with a 0.0005 learning rate, and 10 training epochs. The batch size Bs is 128, window size ws is 16, and multilevel noises $\sigma_{0,1,2,3}$ are set to 0.1, 0.2, 0.3, and 0.4, respectively. Considering the computational load, only 10 000 samples are randomly selected from the unlabeled samples. All experiments are repeated seven times using random seeds, and the median prediction is taken as the final experimental result. Overall accuracy (OA), Kappa, Precision, Recall, $F1$Score coefficients are employed to evaluate the accuracy of change map each comparison method. Equations (12)–(16) provide how OA, Kappa, Precision, Recall, $F1$Score are calculated:

$$\mathrm{OA} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}} \tag{12}$$

$$\mathrm{Kappa} = \frac{\mathrm{OA}-\mathrm{Pre}}{1-\mathrm{Pre}} \tag{13}$$

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \tag{14}$$

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \tag{15}$$

$$F1\mathrm{Score} = 2 \cdot \frac{\mathrm{Precision} \cdot \mathrm{Recall}}{\mathrm{Precision}+\mathrm{Recall}} \tag{16}$$

where TP, TN, FP, and FN are true positive, true negative, false positive, and false negative, respectively. The intermediate value Pre is calculated as follows:

$$\mathrm{Pre} = \frac{(\mathrm{TP}+\mathrm{FP})(\mathrm{TP}+\mathrm{FN})+(\mathrm{TN}+\mathrm{FP})(\mathrm{TN}+\mathrm{FN})}{(\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN})^2}. \tag{17}$$

B. Experimental Results and Analysis on the River Dataset

Table II shows the specific comparison results with a series of recent state-of-the-art (SOTA) methods, including

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TABLE II
RESULT COMPARISON OF THE RIVER EXPERIMENT

	CSDBF 2022	DA-Former 2023	CNCMN 2022	MSDFFN 2023	GTMSiam 2023	ML-EDAN 2022	C3SelF
OA	0.9476	0.8486	0.9378	0.9460	<u>0.9507</u>	0.8975	**0.9660** (↑1.53%)
$Kappa$	0.6322	0.4174	<u>0.6556</u>	0.6108	0.6372	0.5399	**0.8013** (↑14.57%)
$Precision$	0.7520	-	0.6086	<u>0.7629</u>	**0.8187**	-	0.7576 (↓4.13%)
$Recall$	0.6300	-	<u>0.7989</u>	0.5499	0.5573	-	**0.8952** (↑9.63%)
$F1Score$	0.6856	-	<u>0.6909</u>	0.6391	0.6632	-	**0.8207** (↑12.98%)

ML-EDAN [8], MSDFFN [9], DA-Former [14], CSDBF [15], CNCMN [31], and GTMSiam [32]. The optimal detection results are marked in bold, and the suboptimal results are marked in underlines. It can be seen that except for DA-Former and ML-EDAN, the OA values of the remaining comparison methods reach above 0.93 and the Kappa values reach above 0.61, indicating that DA-Former and ML-EDAN are not easy to obtain superior detection results on the River dataset with fewer samples. GTMSiam achieves the suboptimal OA value, and CNCMN achieves the suboptimal Kappa value. C3SelF achieves optimal values in both indicators, especially reflected in the Kappa indicator, which increased by 14.57%. The Precision rate of C3Self is 0.7576, which is slightly lower than the 0.8187 of GTMSiam. The Recall rate of C3Self reaches 0.8952, which is significantly higher than that of other models, which indicates that C3Self is better able to find all relevant positive examples on this dataset. The $F1$Score value of C3Self is 0.8207, which is a better overall performance than the other compared models.

Fig. 4 shows each comparison method change map on the River dataset. There are many false alarms in the detection change map of DA-Former. The prediction change maps of graph-based methods CSDBF and CNCMN are unsatisfactory, indicating that such methods have limited capability to detect scatter-shaped change regions. There are many missed alarms in the change map of GTMSiam, which loses a lot of change information, especially in the lower-left corner. The prediction result of MSDFFN lacks a large amount of detailed information. C3SelF has the highest consistency between the change map and GT. It not only detects a large number of scatter-shaped change regions but also accurately captures the edge and shape information of the change regions.

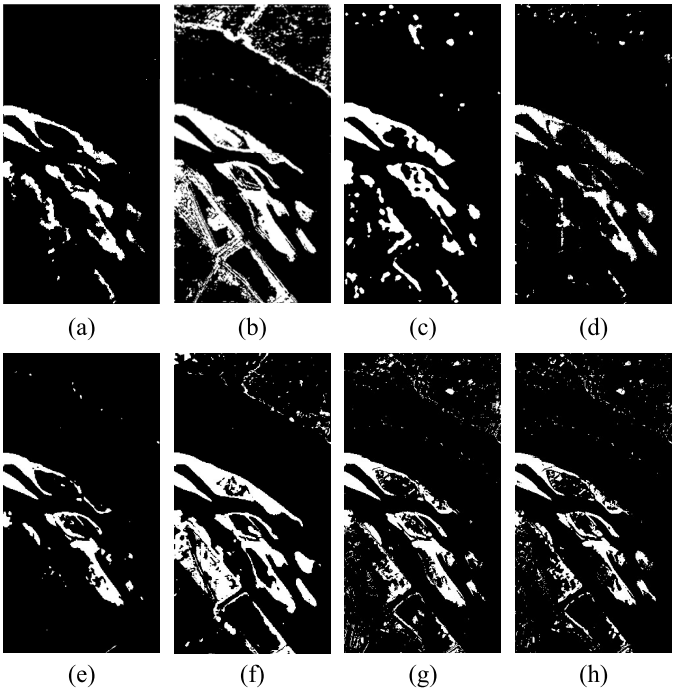


Fig. 4. Detection results of the River dataset with a 0.1% sampling rate. (a) CSDBF. (b) DA-Former. (c) CNCMN. (d) MSDFFN. (e) GTMSiam. (f) ML-EDAN. (g) C3SelF. (h) GT.

C. Experimental Results and Analysis on the Farmland Dataset

From the comparison in Table III, except for DA-Former and ML-EDAN, the other comparison methods obtain similar detection results, with OA values above 0.95 and the Kappa values above 0.88. MSDFFN obtains suboptimal detection results, and C3SelF gets the optimal detection results. Compared with the improvement of the OA, the Kappa improved more, increasing by 1.49% and 3.66%, respectively. The Precision rate of C3Self is 0.7576, which is slightly lower than GTMSiam's 0.8187, but higher than the other models. The Recall rate of C3Self reaches 0.8952, which is significantly higher than that of other models. The $F1$Score value of 0.8952 is significantly higher than other models, which means that C3Self can better find all relevant positive instances.

The change maps of the Farmland dataset are provided in Fig. 5. It can be clearly observed that salt-and-pepper noise exists in the background of the DA-Former change map, and the edges are not clear; ML-EDAN lacks detection of change information and there are some false alarms in the background. Obvious false alarms can also be seen in the detection results of CSDBF and GTMSiam. CNCMN, MSDFFN, and GTMSiam all lack fine discrimination at the edges. The consistency between the change map of C3SelF and the GT map is the highest, and the change information can be identified more accurately.

D. Experimental Results and Analysis on the USA Dataset

The quantitative comparison results of the USA dataset are shown in Table IV. There is a large gap between the comparison methods. The Kappa values of CSDBF, DA-Former,

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TABLE III
RESULT COMPARISON OF THE FARMLAND EXPERIMENT

	CSDBF 2022	DA-Former 2023	CNCMN 2022	MSDFFN 2023	GTMSiam 2023	ML-EDAN 2022	C3SelF
OA	0.9513	0.9453	0.9514	0.9577	0.9575	0.9427	**0.9726** (↑1.49%)
$Kappa$	0.8824	0.8680	0.8840	0.8978	0.8970	0.8602	**0.9344** (↑3.66%)
$Precision$	0.9671	-	0.9696	**0.9756**	0.9734	-	0.9692 (↓0.64%)
$Recall$	0.8946	-	0.9062	0.8968	0.8795	-	**0.9414** (↑3.52%)
$F1Score$	0.9294	-	0.9368	0.9345	0.9241	-	**0.9551** (↑1.83%)

TABLE IV
RESULT COMPARISON OF THE USA EXPERIMENT

	CSDBF 2022	DA-Former 2023	CNCMN 2022	MSDFFN 2023	GTMSiam 2023	ML-EDAN 2022	C3SelF
OA	0.8824	0.8449	0.9272	0.9250	0.9438	0.8434	**0.9481** (↑0.43%)
$Kappa$	0.6320	0.6242	0.7904	0.7756	0.8286	0.5078	**0.8514** (↑2.28%)
$Precision$	0.8152	-	0.8420	0.8791	**0.9538**	-	0.8846 (↓6.92%)
$Recall$	0.6188	-	0.8333	0.7735	0.7892	-	**0.8854** (↑5.21%)
$F1Score$	0.7035	-	0.8376	0.8637	0.6152	-	**0.8850** (↑2.13%)

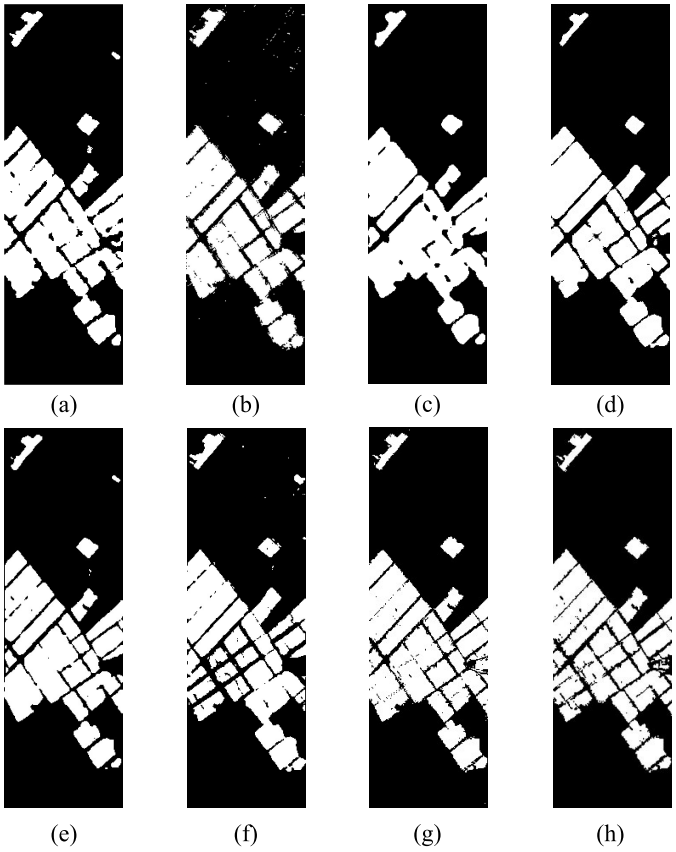


Fig. 5. Detection results of the Farmland dataset with a 0.1% sampling rate. (a) CSDBF. (b) DA-Former. (c) CNCMN. (d) MSDFFN. (e) GTMSiam. (f) ML-EDAN. (g) C3SelF. (h) GT.

and ML-EDAN are below 0.7, and even the Kappa value of ML-EDAN is only 0.5078, which is quite different from the other two; the Kappa values of CNCMN and MSDFFN are

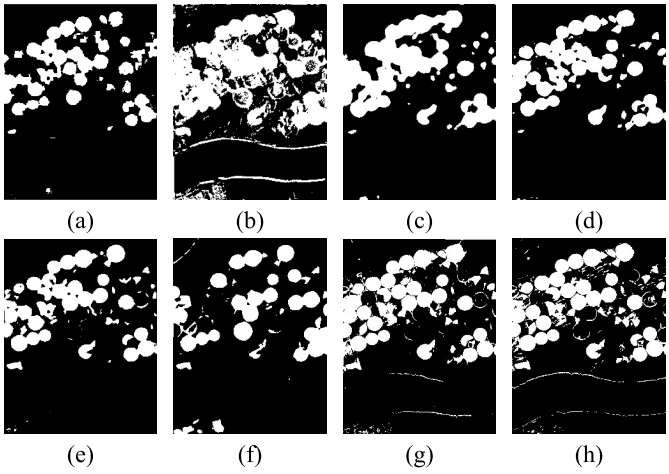


Fig. 6. Detection results of the USA dataset with a 0.1% sampling rate. (a) CSDBF. (b) DA-Former. (c) CNCMN. (d) MSDFFN. (e) GTMSiam. (f) ML-EDAN. (g) C3SelF. (h) GT.

below 0.8, and GTMSiam achieves the suboptimal detection results with the Kappa value of 0.8286, and C3SelF obtains the optimal detection result. Compared with the improvement of the OA, the Kappa increases more, reaching 0.8514. The Precision rate of C3Self is 0.8846, which is lower than that of GTMSiam, but higher than that of CSDBF, CNCMN, and MSDFF. The Recall of C3Self is 0.8854, which is higher than other models except CNCMN, indicating that C3Self can better detect positive cases on this dataset. The $F1$Score value of C3Self is 0.8850, which is higher than other models except CNCMN. This shows that C3Self has a better overall performance on the USA dataset and strikes a better balance between precision and recall.

The visual comparison of the USA dataset is shown in Fig. 6. The differences between the GT map and the change

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TABLE V
ABLATION EXPERIMENTAL SETUP

	Self-ensemble learning	Selective attention mechanism	Multi-level noise perturbation	Class-wise knowledge learning
Baseline	×	×	×	×
SE	✓	×	×	×
Att	✓	×	✓	✓
ML	✓	✓	×	✓
Cls	✓	✓	✓	×
C3SelF	✓	✓	✓	✓

TABLE VI
RESULT COMPARISON OF ABLATION EXPERIMENTS

	River		Farmland		USA	
	OA	$Kappa$	OA	$Kappa$	OA	$Kappa$
Baseline	0.9010	0.3581	0.7900	0.4553	0.8676	0.5521
SE	0.9412	0.7061	0.9509	0.8849	0.9180	0.7546
Att	0.9586	0.7727	0.9651	0.9172	0.9317	0.8014
ML	0.9583	0.7613	0.9634	0.9123	0.9242	0.7861
Cls	0.9616	0.7847	0.9714	0.9310	0.9424	0.8319
C3SelF	**0.9660**	**0.8013**	**0.9726**	**0.9344**	**0.9481**	**0.8514**

TABLE VII
EXPERIMENTS WITH DIFFERENT SAMPLING RATES

	Farmland		River		USA	
Sampling Rates (%)	OA	$Kappa$	OA	$Kappa$	OA	$Kappa$
0.05	0.9541	0.8921	0.9460	0.7203	0.9220	0.7808
0.10	0.9726	0.9344	0.9660	0.8013	0.9481	0.8514
0.15	0.9773	0.9451	0.9700	0.8187	0.9512	0.8584

maps of DA-Former, CNCMN, and ML-EDAN are large, and the visual effect is not satisfactory. CSDBF, MSDFFN, and GTMSiam cannot identify changes at river lines. C3SelF has the highest consistency between the change map and the GT map, the change information can be identified more accurately.

E. Ablation Experiments

To evaluate the role played by each component in the proposed C3SelF framework, a series of ablation experiments are designed in this section. The specific conditions are shown in Table V. "SE" is the experiment with only self-ensemble learning; "Att" is the experiment with ablation of selective attention learning; "ML" is the experiment with ablation of multilevel noise perturbation; 'Cls" is the experiment with ablation of class-wise knowledge learning.

Table VI shows the comparison of each ablation experiment by the objective indicator. It can be clearly observed that each part plays a positive role in the framework. It is difficult for the baseline experiment to learn effective discriminative features, which makes the detection performance of each experiment very bad. Since the addition of SE, the accuracy has increased dramatically, demonstrating that self-integrated learning has better detection performance compared to a single learning model. Adding Att on top of SE realizes focusing on effective feature extraction and processing for complex data, resulting in improved detection accuracy. For the addition of ML, compared to the baseline as well as SE combinations, the results are better than both, indicating a multilevel noise perturbation strategy that enhances the robustness of the network under multiple interferences. With the introduction of class-wise learning, the model detection accuracy is highly improved; it is demonstrated that through class-wise knowledge learning, the model can learn a wider range of class features, which enhances its generalization ability on unlabeled samples, enabling the model to better understand the differences and connections between classes. Finally, C3Self is evaluated to achieve the best metrics and alleviate the problem of fewer samples caused by the difficulty of making labels for changes in HSIs. The rest of the experiments also contributed positively to the precision of the detection results to varying degrees.

Fig. 7 shows the visual comparison of each ablation experiment. There is a lot of noise in the baseline change map, and the visualization effect is very poor. A large amount of false alarms exist in the SE experimental results, especially for the USA dataset, the detection result is not satisfactory. The detection results of other ablation experiments have been improved to a certain extent compared with SE, but they are still not excellent enough in terms of edge change information discrimination.

After statistics, the parameter of the S3Net framework is 759.07K, and the floating-point operations (FLOPs) are 1.609G. As shown in Table V, the S3Net adds a selective AM and class-wise knowledge learning. The number of parameters increases by 0.03 K compared with the baseline. This means that the S3Net framework does not significantly increase the model size, and the additional parameters will not impose an excessive burden on the hardware resources during model storage and loading. The FLOPs increase by 0.001G, which indicates that the computational complexity of S3Net is almost the same as that of the baseline model. Under the same hardware environment, there will be no significant difference in computational resource consumption.

F. Discussion of Hyperparameters

We also experimented on three different datasets for sampling rates of 0.05%, 0.10%, and 0.15%, giving experimental results in Table VII, as shown in Fig. 8.

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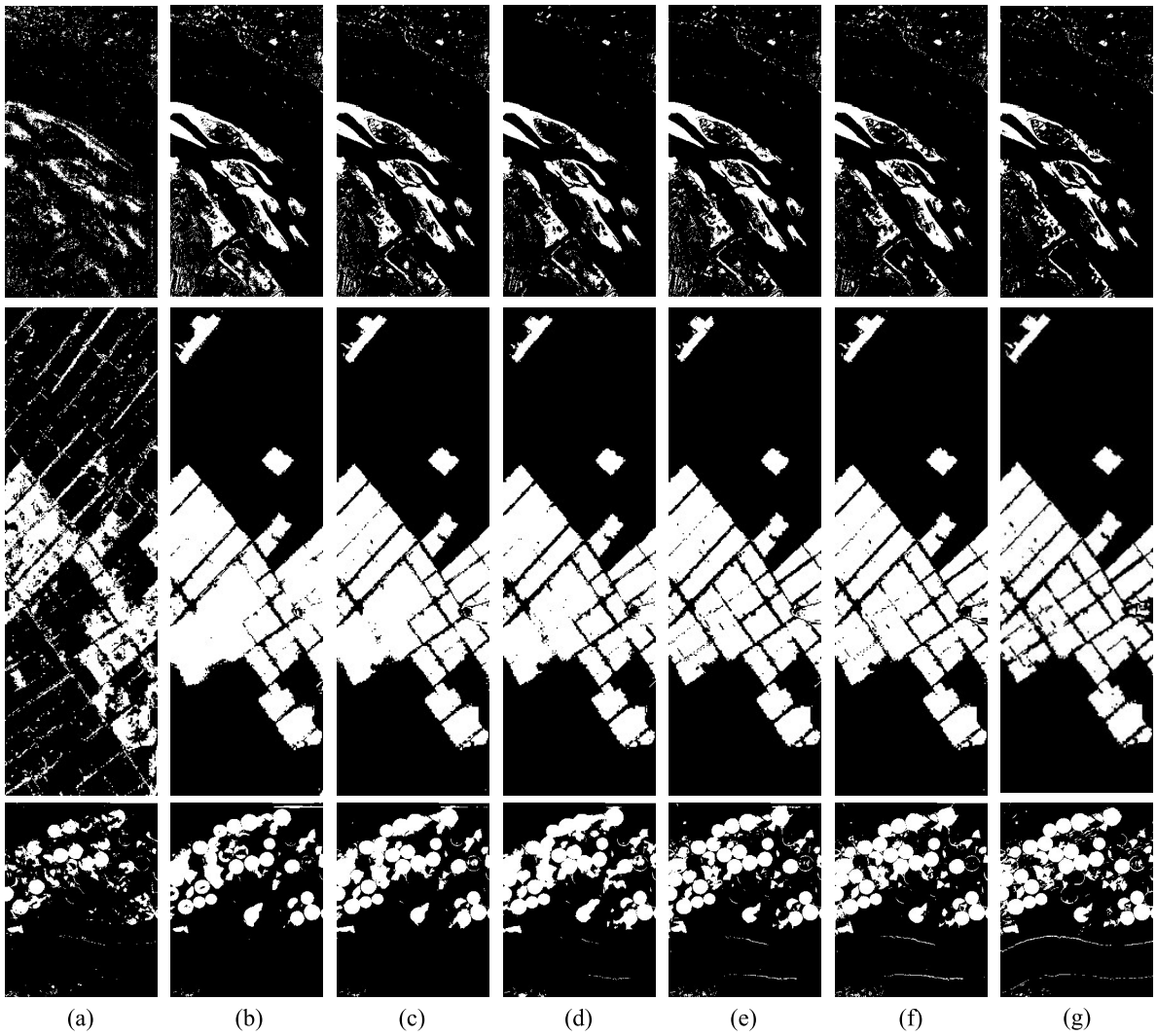


Fig. 7. Comparison of ablation experimental results. (a) Baseline. (b) SE. (c) Att. (d) ML. (e) Cls. (f) C3SelF. (g) GT.

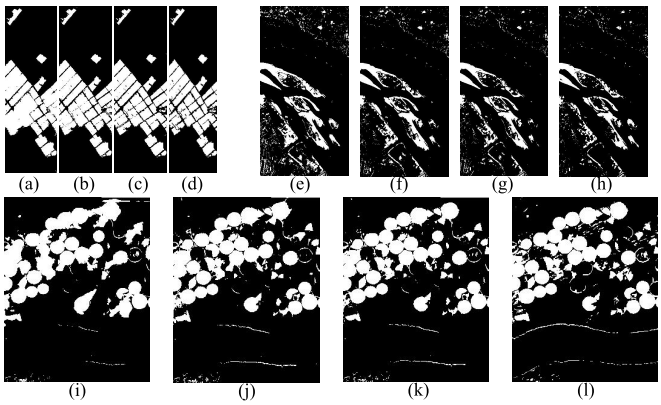


Fig. 8. Experimental results for multiple sampling rates. (a) Sampling rate at 0.05% of the River dataset. (b) Sampling rate at 0.10% of the River dataset. (c) Sampling rate at 0.15% of the River dataset. (d) GT of the River dataset. (e) Sampling rate at 0.05% of the Farmland dataset. (f) Sampling rate at 0.10% of the Farmland dataset. (g) Sampling rate at 0.15% of the Farmland dataset. (h) GT of the Farmland dataset. (i) Sampling rate at 0.05% of the USA dataset. (j) Sampling rate at 0.10% of the USA dataset. (k) Sampling rate at 0.15% of the USA dataset. (l) GT of the USA dataset.

Overall, increasing the sampling rate improves C3self's OA and Kappa for different datasets (Farmland, River, and USA) to a certain extent, but the enhancement effect diminishes as the sampling rate continues to increase. This means that increasing the sampling rate within a certain range has a positive impact on the detection results, and the choice of sampling rate needs to be weighed on a case-by-case basis to achieve a better balance between efficiency and accuracy in practical applications.

Selecting multiple different levels of noise allows for a more comprehensive simulation of these real-world scenarios to test the model's performance in different noise environments and thus assess the robustness of the model. This experiment is shown in Table VIII, as shown in Fig. 9.

These noise levels are chosen, on the one hand, to verify the effect of the multilevel noise perturbation strategy on the performance advantage of the C3Self model and to assess the generalization ability of the model more comprehensively. As can be seen from the table, the model achieves good OA and Kappa values, which means that it can effectively extract and utilize the information in HSIs for CD under different noise conditions. The diversity of the data can be increased by introducing multiple levels of noise interference

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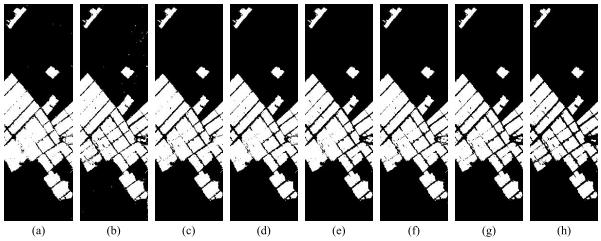


Fig. 9. Multilevel noise perturbation results of the Farmland dataset. (a) σ_0. (b) σ_1. (c) σ_2, (d) σ_3. (e) σ_0 and σ_1. (f) σ_0, σ_1, and σ_2. (g) σ_0, σ_1, and σ_3. (h) σ_1, σ_2, and σ_3.

TABLE VIII
EXPERIMENTS WITH DIFFERENT NOISE DISTURBANCE RATES ON THE FARMLAND DATASET

σ_n	0	1	2	3	OA	$Kappa$
value	0.1	0.2	0.3	0.4		
	✓				0.9632	0.9133
		✓			0.9692	0.9268
			✓		0.9690	0.9260
				✓	0.9686	0.9250
	✓	✓			0.9693	0.9262
	✓		✓		0.9704	0.9291
	✓			✓	0.9700	0.9280
		✓	✓		0.9705	0.9294
		✓		✓	0.9698	0.9279
			✓	✓	0.9712	0.9310
	✓	✓	✓		0.9721	0.9333
	✓	✓		✓	0.9728	0.9342
	✓		✓	✓	0.9729	0.9344
		✓	✓	✓	0.9730	0.9333
	✓	✓	✓	✓	0.9726	0.9344

when the amount of data is limited. This diversity helps the model to learn a wider range of feature representations, thus improving its generalization ability. The introduction of noise forces the model to not only recognize the original signal during training but also learn to distinguish between noise and useful information. This "denoising" process helps the model to develop stronger feature extraction capabilities and to be able to extract key information from complex or ambiguous inputs. Multilevel noise interference exposes the model to more diverse inputs during the training process, which effectively prevents overfitting by enhancing the robustness and generalization of the model.

V. CONCLUSION

The problem of fewer samples caused by the difficulty in producing change labels for bi-temporal HSIs needs to be solved urgently. C3SelF is proposed in this article, aiming to use the SSL strategy to enhance the utilization of valid information from unlabeled samples and reduce the demand for training samples. On this basis, a multilevel noise perturbation and lightweight S3Net are designed to mitigate overfitting. Specifically, the framework consists of two same-structured networks, called CD network and self-ensemble learning network. The constraints include patch-wise classification loss, class-wise consistency loss on labeled samples, and patch-wise consistency loss on unlabeled samples. They are used to enable the CD network to learn effective discriminative features while allowing the self-ensemble learning network to aggregate them, which enhances the robustness and stability of the network for detection under the few-shot condition. Experiments verify the superior detection performance of the proposed framework, which achieves SOTA performance.

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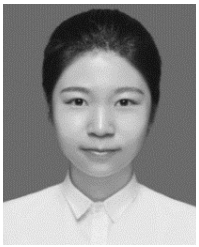
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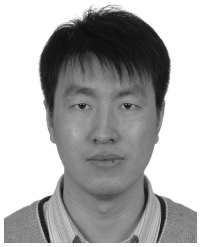
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